# Free-body diagrams

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Draw only the external forces on one chosen object.

**SUCCESS CRITERIA - I can...**

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “normal force” and “tension” to explain your model or result. Sentence starters are optional.

**normal force**: a support force perpendicular to a contact surface

**tension**: the pulling force through a stretched rope or cable

**weight**: the gravitational force on an object

## 01 Warm-up: bring back an earlier idea

A fixed-mass cart has acceleration 3 m/s² under 6 N net force.
**Find** acceleration when net force becomes 10 N.
**Explain**.

**Word help:** net force = the vector sum of all forces on one object

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = 9.8 N/kg unless another value is given.

**Worked reasoning:** A lamp hangs motionless from one vertical cable. With weight 12 N down, the cable tension is 12 N up. Only forces acting on the lamp belong in its diagram.

**Your turn: Draw** the lamp as a labeled point with its two force arrows.
**Explain** why its force on the cable is not another arrow on this diagram.

**Sentence starter:** My claim is … . The diagram/data show …, so … .



Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 A book at rest

**Draw** and label the forces on a book resting on a level table.
**Name** both sources.
**Use** arrows whose comparative lengths match rest.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .



Add and label force arrows.

The dot represents the object. Object: book

**Marzano check - circle:** ① ② ③

## 04 After the hand lets go

A box slides to the right on a rough floor after release.
**Draw** its forces. Which way does friction act?

**Sentence starter:** The ... in my model represents ... . Its label shows ... .



Add and label force arrows.

Box slides right after the hand lets go.

**Marzano check - circle:** ① ② ③

## 05 A falling ball

A ball falls down and is speeding up. Air pushes up.
**Draw** weight and air resistance. Which arrow should be longer?

**Sentence starter:** The ... in my model represents ... . Its label shows ... .



Add and label force arrows.

Falling ball speeds up; air resistance acts upward.

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A cart goes from 18 m to 6 m in 4 s at steady velocity.
**Find** velocity and sketch its position-time graph.
**Label** the model with the given positions and any prediction.

**Word help:** velocity = rate of position change, with direction

**Sentence starter:** I predict ... . My model supports this because ... .

Position (m)



Time (s)

**Marzano check - circle:** ① ② ③

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Sentence starters are optional. You may plan in any language.

## 07 Exit check

A ball moves upward after leaving a hand. Ignore air resistance.
**Draw** its free-body diagram and justify every arrow.

**Word help:** justify = support your answer with a reason or evidence

**Sentence starter:** My claim is … . The diagram/data show …, so … .

Add and label force arrows.

Ball travels upward after leaving the hand.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

**Draw** separate diagrams for a hanging lamp at rest and the same lamp moving upward at steady speed. What can a force diagram alone fail to tell you?

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** The … in my model represents … . Its label shows … .

Lamp at rest

Lamp moving up steadily

**Marzano check - circle:** 1 2 3